

Two-Step Equations & Checking Solutions

solving two-step equations, including negative coefficients, and checking solutions by substitution

The pattern (worked for you): to solve $3x + 4 = 10$, undo the $+4$ first (subtract 4 from both sides: $3x = 6$), then undo the $\times 3$ (divide both sides by 3: $x = 2$). *Check* by substituting back: $3(2) + 4 = 10$. ✓

- Undo addition or subtraction first, then multiplication or division.
- Check every solution by substituting it into the original equation.

1. Solve and check: $2x + 3 = 11$

Answer: _____

2. Solve and check: $3x - 5 = 16$

Answer: _____

3. Solve and check: $\frac{x}{4} + 2 = 7$

Answer: _____

4. Solve and check: $5x + 9 = -6$

Answer: _____

5. Solve and check: $\frac{x}{3} - 8 = -2$

Answer: _____

6. Solve and check: $10 - 2x = 4$

Answer: _____

7. Nina says that $x = 6$ is the solution of $4x - 7 = 17$.

(a) Check her claim: evaluate $4x - 7$ when $x = 6$: _____

(b) Is Nina correct? Say how your value from part (a) tells you.

8. Solve and check: $7 = 2x - 9$

Answer: _____

9. Solve and check: $\frac{x+5}{3} = 4$

Answer: _____

10. Solve and check: $-4x + 3 = 19$

Answer: _____

11. Leo solves $4x - 6 = 18$ and gets $x = 3$.

(a) Check Leo's answer: evaluate $4x - 6$ when $x = 3$: _____

(b) Solve the equation correctly.

Answer: _____

(c) What error did Leo make? Answer in one sentence.

12. Challenge. A bike rental costs a flat fee of 4 dollars plus 2.50 dollars per hour. Amara paid 16.50 dollars in total, so the number of hours h she rented satisfies

$$2.5h + 4 = 16.5.$$

Solve the equation for h , and check your solution.

Answer: _____